INFO8006 Introduction to Artificial Intelligence

Exercise session 4: Probabilistic reasoning

Recap

Bayesian networks

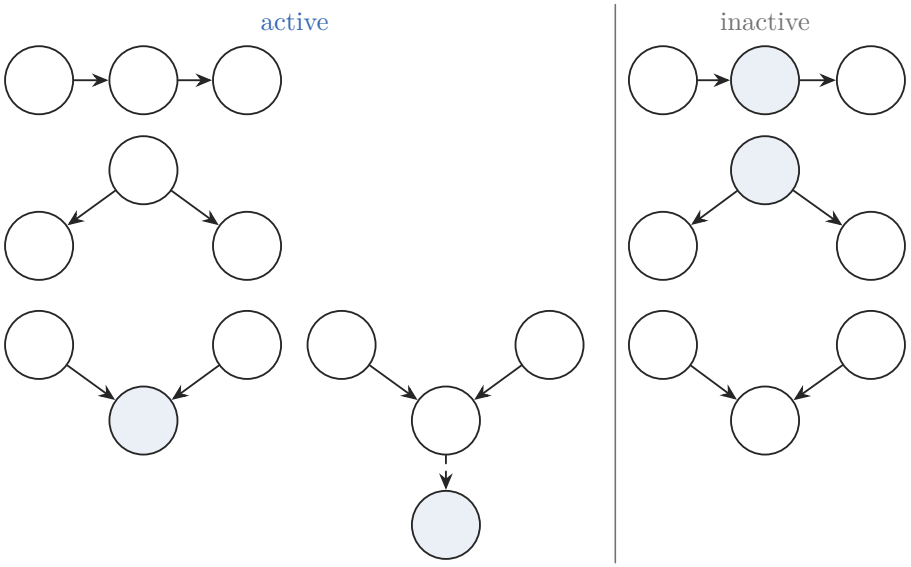
The state s and the percepts e of an agent are assignments of random variables $X_1, \ldots, X_n$, with joint distribution $P(s, e) = P(s)P(e \mid s)$. A *Bayesian network* is a directed acyclic graph with one node per variable X_i, annotated with a *conditional probability table* (CPT) $\mathbf{P}(X_i \mid \text{parents}(X_i))$. It asserts the *local Markov property* $X_i \perp \text{non-descendants}(X_i) \mid \text{parents}(X_i)$, hence

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid \text{parents}(X_i)).$$

d-separation

d-separation reads independences from the structure alone: $X \perp Y \mid \mathbf{Z}$ holds if every undirected path between X and Y is *inactive*. If a path is *active*, nothing follows: the independence may still hold, through the values of the CPTs. A path is active iff all its triples are:

Triple		Active iff
Cascade	$A \to B \to C$ or $A \leftarrow B \leftarrow C$	B unobserved
Common parent	$A \leftarrow B \to C$	B unobserved
v-structure	$A \to B \leftarrow C$	B or a descendant of B observed



Given its *Markov blanket* (parents, children, other parents of its children), a variable is independent of all others.

Inference (optional)

For a *query* Q, *evidence* $\mathbf{E} = \mathbf{e}$ and *hidden* variables $\mathbf{H}$,

$$\mathbf{P}(Q \mid \mathbf{e}) = \frac{1}{Z} \sum_{\mathbf{h}} \mathbf{P}(Q, \mathbf{h}, \mathbf{e}), \qquad Z = \sum_{q} \sum_{\mathbf{h}} P(q, \mathbf{h}, \mathbf{e}).$$

- *Enumeration* evaluates the sums depth-first: space $O(n)$, time $O(d^n)$ for n variables of d values.
- *Variable elimination joins factors*, initially the CPTs given $\mathbf{e}$, by pointwise products, and *eliminates* hidden variables by summing over their values. Its cost, that of the largest factor, depends on the ordering; the best one is NP-hard to find.

Parameter learning

For a parameterized family $\mathbf{P}(X \mid \theta)$ and i.i.d. data $\mathbf{d} = \{x_1, \ldots, x_N\}$:

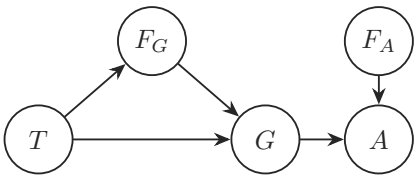
Likelihood	$P(\mathbf{d} \mid \theta) = \prod_{j=1}^{N} P(x_j \mid \theta), \quad L(\theta) = \log P(\mathbf{d} \mid \theta)$
Maximum likelihood (MLE)	$\theta^*_{\text{MLE}} = \arg\max_\theta L(\theta)$, often from $\partial L / \partial \theta = 0$
Posterior	$\mathbf{P}(\theta \mid \mathbf{d}) \propto \mathbf{P}(\mathbf{d} \mid \theta)\mathbf{P}(\theta)$, for a prior $\mathbf{P}(\theta)$
Maximum a posteriori (MAP)	$\theta^*_{\text{MAP}} = \arg\max_\theta P(\mathbf{d} \mid \theta)P(\theta)$

In session: exercises 1, 2.
Recommended before the next session: exercise 5.

Exercise 1 Nuclear power plant (AIMA ch. 13)
A monitoring program supervises the core of a nuclear power station. An alarm sounds when a temperature gauge, which measures the temperature of the core, exceeds a given threshold, and the program reasons about the actual temperature from the gauge and the alarm, either of which may be faulty. Consider the Boolean variables A (alarm sounds), F_A (alarm is faulty), and F_G (gauge is faulty) and the multi-valued nodes G (gauge reading) and T (actual core temperature).

(a) Draw a Bayesian network for this domain, given that the gauge is more likely to fail when the core temperature gets too high.

Solution.



(b) From your network topology, can you conclude $T \perp F_A \mid F_G$ and $T \perp F_A \mid F_G, A$?

Solution. The paths from T to F_A are (T, F_G, G, A, F_A) and (T, G, A, F_A).
Given F_G, the triple (G, A, F_A) is inactive (v-structure, A unobserved and without descendants). It lies on both paths, so $T \perp F_A \mid F_G$.
Given F_G and A, the triple (G, A, F_A) becomes active. The first path is still blocked by (T, F_G, G) (cascade, F_G observed), but on the second path (T, G, A) (cascade, G unobserved) and (G, A, F_A) are both active. We cannot conclude $T \perp F_A \mid F_G, A$.

(c) Suppose there are just two possible actual and measured temperatures: low (l) and high (h). The probability that the gauge gives the correct temperature is x when it is working, but y when it is faulty. Give the conditional probability table associated with G.

Solution. The CPT for G is shown below. Note the semantics of F_G, which is true when the gauge is faulty, *i.e.* not working properly.

T	F_G	$P(G = l \mid T, F_G)$
l	0	x
l	1	y
h	0	$1 - x$
h	1	$1 - y$

(d) Suppose the alarm is always triggered by high measured temperatures, unless it is faulty, in which case it never sounds. Give the conditional probability table associated with A.

Solution.

G	F_A	$P(A = 1 \mid G, F_A)$
l	0	0
l	1	0
h	0	1
h	1	0

(e) Suppose the gauge is not faulty and the alarm is triggered. Calculate an expression for the probability that the temperature of the core is too high, in terms of the various conditional probabilities in the network.

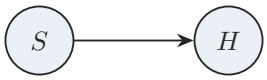
Solution. The probability of interest here is $P(T = h \mid A = 1, F_G = 0)$. Because the alarm's behavior is deterministic, we can deduce that G indicates high (h). Hence, $P(T = h \mid A = 1, F_G = 0) = P(T = h \mid A = 1, G = h, F_G = 0)$. Every path from A to T goes through G, which is observed and is not the center of a v-structure on these paths, so $A \perp T \mid G, F_G$. Therefore, we only need to calculate

$$\begin{aligned} P(T = h \mid G = h, F_G = 0) &= \frac{P(T = h, G = h, F_G = 0)}{P(G = h, F_G = 0)} \\ &= \frac{P(G = h \mid T = h, F_G = 0)P(F_G = 0 \mid T = h)P(T = h)}{\sum_t P(G = h \mid t, F_G = 0)P(F_G = 0 \mid t)P(t)} \\ &= \frac{x\pi_h}{x\pi_h + (1 - x)\pi_l}, \end{aligned}$$

with $\pi_t = P(F_G = 0 \mid T = t)P(T = t)$, using the CPT of (c), $P(G = h \mid T = h, F_G = 0) = x$ and $P(G = h \mid T = l, F_G = 0) = 1 - x$. It cannot be developed further, as $P(T)$ and $P(F_G \mid T)$ are not given.

Exercise 2 Energy of an inference server

A scheduling program manages a data center in which a GPU server answers the requests sent to a language model. To plan the power supply, the scheduler models the energy the server consumes as a function of its load, with the following network.



where S is a random variable denoting the number of requests the server receives in an hour, and H a random variable denoting the energy, in kilowatt-hours, that it consumes during that hour. The scheduler chooses the following relations for each variable:

$$P(S = s) = \texttt{Poisson}(s \mid \lambda) = \frac{\lambda^s e^{-\lambda}}{s!} \tag{1}$$

$$H = \omega S + H_0 + \mathcal{N}(0, \sigma^2) \tag{2}$$

where H_0 is the energy the server consumes when idle, which is known, and ω is the energy consumed per request. The term $\mathcal{N}(0, \sigma^2)$ is a Gaussian noise of mean 0 and variance σ^2, where the Gaussian density of mean μ and variance σ^2 is

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

(a) Identify the parameters of the Bayesian network and give the joint log-likelihood of one pair (s_i, h_i) given the model.

Solution. Given the topology of the network, we have to identify $P(S = s)$ and $P(H = h \mid S = s)$. The former is given by (1) whereas the latter can be deduced from (2). We have

$$P(S = s) = \texttt{Poisson}(s \mid \lambda)$$

and

$$p(h \mid s) = \mathcal{N}(h \mid \omega s + H_0, \sigma^2).$$

The parameters of those distributions are $\theta = (\lambda, \omega, \sigma)$.
The likelihood of a pair (s_i, h_i) can be derived from the network. We have

$$\begin{aligned} p((s_i, h_i) \mid \theta) &= P(s_i)\, p(h_i \mid s_i) \\ &= \frac{\lambda^{s_i} e^{-\lambda}}{s_i!} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(h_i - \omega s_i - H_0)^2}{2\sigma^2}} \end{aligned}$$

from which we derive the log-likelihood

$$L((s_i, h_i); \theta) = s_i \log \lambda - \lambda - \log s_i! - \log \sqrt{2\pi}\sigma - \frac{(h_i - \omega s_i - H_0)^2}{2\sigma^2}. \tag{3}$$

(b) The scheduler collects a set $\mathbf{d} = \{(s_i, h_i)\}_{i=1}^N$ of independent observations. Derive the maximum likelihood estimate of the parameters.

Solution. We first have to define the joint likelihood of those observations. Since they are assumed i.i.d., we have

$$p(\mathbf{d} \mid \theta) = \prod_{i=1}^{N} p((s_i, h_i) \mid \theta)$$

and the log-likelihood is

$$\begin{aligned} L(\mathbf{d}; \theta) &= \sum_i L((s_i, h_i); \theta) \\ &= -N(\lambda + \log \sqrt{2\pi}\sigma) + \sum_i \left(s_i \log \lambda - \log s_i! - \frac{(h_i - \omega s_i - H_0)^2}{2\sigma^2} \right) \end{aligned}$$

The maximum likelihood parameters can be found by setting to zero the derivative of the log-likelihood w.r.t. each parameter.
For λ we have

$$\begin{aligned} \frac{\partial L}{\partial \lambda} &= -N + \sum_i \frac{s_i}{\lambda} \\ &= 0 \end{aligned}$$

which gives $\boxed{\lambda_{\text{MLE}} = \frac{1}{N} \sum_i s_i}$.
For ω, whose estimate does not depend on σ, we have

$$\begin{aligned} \frac{\partial L}{\partial \omega} &= \sum_i \frac{s_i(h_i - \omega s_i - H_0)}{\sigma^2} \\ &\propto \left(\sum_i s_i(h_i - H_0) \right) - \omega \sum_i s_i^2 \\ &= 0 \end{aligned}$$

which gives $\boxed{\omega_{\text{MLE}} = \frac{\sum_i s_i(h_i - H_0)}{\sum_i s_i^2}}$.
For σ we have

$$\begin{aligned} \frac{\partial L}{\partial \sigma} &= -\frac{1}{\sigma} \left(N - \sum_i \frac{(h_i - \omega s_i - H_0)^2}{\sigma^2} \right) \\ &\propto N - \frac{1}{\sigma^2} \sum_i (h_i - \omega s_i - H_0)^2 \\ &= 0 \end{aligned}$$

which gives, with $\omega = \omega_{\text{MLE}}$, $\boxed{\sigma^2_{\text{MLE}} = \frac{1}{N}\sum_i (h_i - \omega_{\text{MLE}} s_i - H_0)^2}$.

(c) The scheduler has just started to collect observations, and maximum likelihood can overfit when they are few. It therefore treats the parameter λ, about which it has no strong knowledge, as uncertain. Using a prior $\lambda \sim \texttt{Gamma}(\alpha, \beta)$, what would be the posterior distribution $p(\lambda \mid \mathbf{s})$, with $\mathbf{s} = \{s_i\}_{i=1}^N$?
Hint: $\texttt{Gamma}(x \mid \alpha, \beta) = \frac{1}{Z} x^{\alpha-1} e^{-\beta x}$, with Z a normalizing constant.

Solution. Using Bayes' rule

$$\begin{aligned} p(\lambda \mid \mathbf{s}) &\propto P(\mathbf{s} \mid \lambda) p(\lambda) \\ &\propto \lambda^{\alpha-1} e^{-\beta\lambda} \prod_i \frac{\lambda^{s_i} e^{-\lambda}}{s_i!} \\ &= \lambda^{\alpha-1} e^{-\beta\lambda} \left(\frac{\lambda^{\sum_i s_i} e^{-N\lambda}}{\prod_i s_i!} \right) \\ &\propto \lambda^{(\alpha + \sum_i s_i - 1)} e^{-(\beta+N)\lambda} \end{aligned}$$

which corresponds to $\texttt{Gamma}(\alpha + \sum_i s_i,\ \beta + N)$, the posterior density. The symbol $\propto$ means equality up to a normalizing constant: every factor that does not depend on λ can be absorbed in it.

(d) Knowing that the mean, the variance and the mode of $\texttt{Gamma}(\alpha, \beta)$ are respectively $\frac{\alpha}{\beta}$, $\frac{\alpha}{\beta^2}$ and $\frac{\alpha - 1}{\beta}$, interpret how they evolve w.r.t. N between the prior and the posterior.

Solution. We have

$$\mathbb{E}(\lambda) : \frac{\alpha}{\beta} \Rightarrow \frac{\alpha + \sum_i s_i}{\beta + N},$$

$$\mathbb{V}(\lambda) : \frac{\alpha}{\beta^2} \Rightarrow \frac{\alpha + \sum_i s_i}{(\beta + N)^2}$$

and

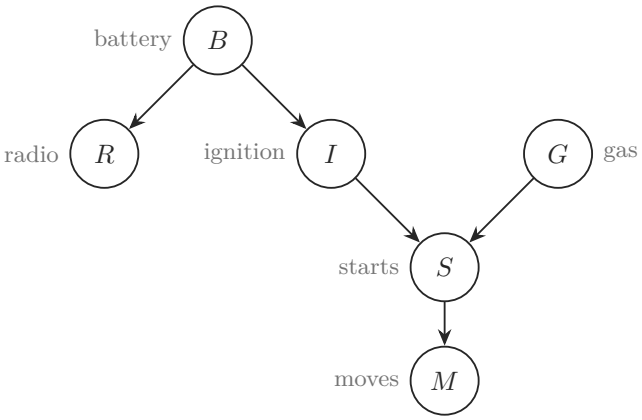
$$\lambda_{\text{MAP}} = \arg\max_\lambda p(\lambda \mid \mathbf{s}) : \frac{\alpha - 1}{\beta} \Rightarrow \frac{\alpha + \sum_i s_i - 1}{\beta + N}.$$

The mode of the posterior is the MAP estimate λ_{MAP}. When $N = 0$, the prior and the posterior are identical, as expected. When $N \to \infty$, $\sum_i s_i \approx N\lambda$ grows linearly with N, so $\sum_i s_i \gg \alpha$ and $N \gg \beta$: the mean and the MAP estimate converge to $\lambda_{\text{MLE}} = \frac{1}{N}\sum_i s_i$, and the variance, about λ_{MLE}/N, goes to 0, so that the posterior concentrates on λ_{MLE}. The posterior mean

$$\frac{\alpha + \sum_i s_i}{\beta + N} = \frac{\beta}{\beta + N}\frac{\alpha}{\beta} + \frac{N}{\beta + N}\lambda_{\text{MLE}}$$

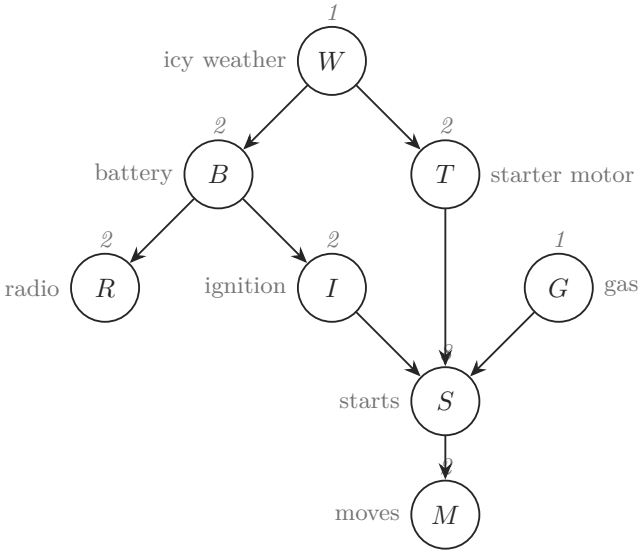
moves from the prior mean to the MLE as data accumulate.

Exercise 3 Car diagnosis (AIMA ch. 13)
Consider the following Bayesian network, which describes some features of a car's electrical system and engine. Each variable is Boolean, and the true value indicates that the corresponding aspect of the vehicle is in working order.



(a) Extend the network with the Boolean variables *IcyWeather* and *StarterMotor*.

Solution. IcyWeather is not caused by any of the car-related variables, so needs no parents. It directly affects the battery and the starter motor. StarterMotor is an additional precondition for Starts.



(b) According to your knowledge of cars, give reasonable conditional probability tables for all the nodes.

Solution. Reasonable probabilities may vary a lot depending on the kind of car and perhaps the personal experience of the assessor. The following values indicate the general order of magnitude and relative values that would be reasonable:

- A reasonable prior for IcyWeather might be 0.05 (depending on location and season).
- $P(Battery \mid IcyWeather) = 0.95, P(Battery \mid \neg IcyWeather) = 0.997$.
- $P(StarterMotor \mid IcyWeather) = 0.98,\ P(StarterMotor \mid \neg IcyWeather) = 0.999$.
- $P(Radio \mid Battery) = 0.9999, P(Radio \mid \neg Battery) = 0.05$.
- $P(Ignition \mid Battery) = 0.998, P(Ignition \mid \neg Battery) = 0.01$.
- $P(Gas) = 0.995$.
- $P(Starts \mid Ignition, StarterMotor, Gas) = 0.9999$, other entries 0.0.
- $P(Moves \mid Starts) = 0.998,\ P(Moves \mid \neg Starts) = 0$.

(c) How many independent values are contained in the joint probability distribution for eight Boolean nodes, assuming that no conditional independence relations are known to hold among them?

Solution. With 8 Boolean variables, the joint has $2^8 - 1 = 255$ independent entries.

(d) How many independent probability values do your network tables contain?

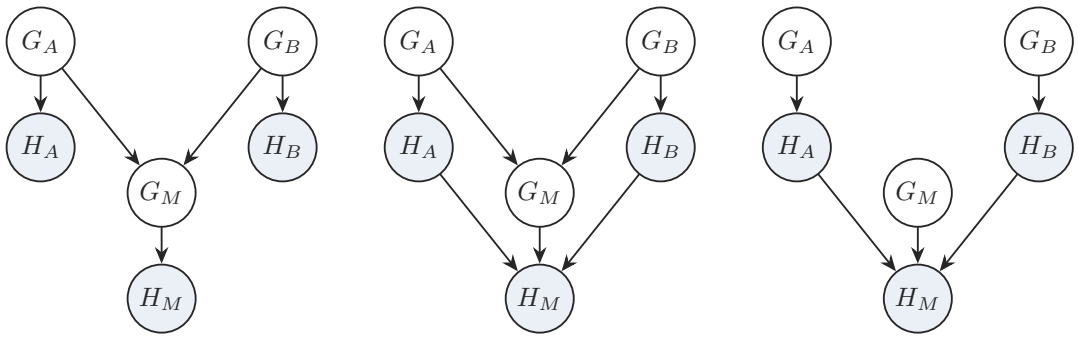
Solution. Given the topology with IcyWeather and StarterMotor, the total number of independent CPT entries is $1 + 2 + 2 + 2 + 2 + 1 + 8 + 2 = 20$.

Exercise 4 Evolutionary model merging (AIMA ch. 13)

Evolutionary model merging builds a new language model M from two parent models A and B: each component of M is copied from one of the parents. Consider one component, and let G_x be the language it was mostly trained on in model x, with possible values e (English) or f (French). A common *hypothesis* is that the default language of a model follows from this component by a simple mechanism: the language H_x in which model x answers by default is the same as G_x with some probability s.

Furthermore, M copies the component from either parent with equal probability, and a random perturbation flips it with a small nonzero probability m.

(a) Which of the following networks claim that $\mathbf{P}(G_M \mid G_A, G_B) = \mathbf{P}(G_M)$?



Solution. The third one: G_M has no parents, so by the local Markov property it is independent of its non-descendants G_A and G_B.

(b) Which of the networks make independence claims that are consistent with the hypothesis about the default language?

Solution. The first and second ones. The second one is also valid as it makes *fewer* assumptions about the dependencies between the variables, *i.e.* it is more general than the first one. The third one makes G_M independent of G_A and G_B, which contradicts the copying of the component from a parent.

(c) Write down the CPT for the G_M node in the first network, in terms of s and m.

Solution.

g_A	g_B	$P(G_M = f \mid g_A, g_B)$
f	f	$1 - m$
f	e	0.5
e	f	0.5
e	e	m

The CPT does not depend on s, which only relates G_M and H_M.

(d) Suppose that $P(G_A = f) = P(G_B = f) = q$. In the first network, derive an expression for $P(G_M = f)$ in terms of s, m and q.

Solution.

$$\begin{aligned} P(G_M = f) &= \sum_{g_A} \sum_{g_B} P(G_M = f, g_A, g_B) \\ &= \sum_{g_A} \sum_{g_B} P(G_M = f \mid g_A, g_B) P(g_A) P(g_B) \\ &= (1-m)q^2 + 0.5(1-q)q + 0.5q(1-q) + m(1-q)^2 \\ &= q + m - 2qm \end{aligned}$$

Again, s does not appear.

(e) Over many generations of merges, we expect the distribution of the component to be the same in the parents and in the merged models. Use this to calculate the value of q. Only about 10 % of the merged models answer in French by default: explain why the hypothesis described at the beginning of this question must be wrong.

Solution. At equilibrium, $P(G_M = f) = P(G_A = f) = q$, so $q = q + m - 2qm$, *i.e.* $m(1-2q) = 0$. Since $m > 0$, $q = 0.5$. Then $P(H_M = f) = sq + (1-s)(1-q) = 0.5$ whatever s: half of the merged models should answer in French, against about 10 % observed, which invalidates the hypothesis.

Exercise 5 Lidar calibration

A self-driving car calibrates its new lidar against a reference target before driving. The distance μ_y at which the lidar sees the target is uncertain: the specification of the calibration station gives the prior $\mu_y \sim \mathcal{N}(\mu_l, \sigma_l^2)$. Each reading y_i of the lidar is Gaussian, $y_i \sim \mathcal{N}(\mu_y, \sigma^2)$, with a noise variance σ^2 known from its datasheet.

The calibration program of the car records a calibration set $\mathcal{Y}_c$ of C readings, and wants to predict the next reading y of the lidar. We study how much $\mathcal{Y}_c$ reduces the uncertainty about this reading. Before the calibration, the reading is predicted by the *prior predictive* distribution $p(y) = \int p(y \mid \mu_y) p(\mu_y)\, d\mu_y$; after it, by the *posterior predictive* distribution $p(y \mid \mathcal{Y}_c) = \int p(y \mid \mu_y) p(\mu_y \mid \mathcal{Y}_c)\, d\mu_y$.

You may use the following identities. If $p(x) = \mathcal{N}(x \mid m, S)$ and $p(z \mid x) = \mathcal{N}(z \mid hx + u, R)$, then x and z are jointly Gaussian,

$$p(x, z) = \mathcal{N}\left(\begin{pmatrix} x \\ z \end{pmatrix} \middle| \begin{pmatrix} m \\ hm + u \end{pmatrix}, \begin{pmatrix} S & hS \\ hS & h^2 S + R \end{pmatrix} \right).$$

If x and z are jointly Gaussian with means a and b, variances A and B, and covariance K, then

$$p(z) = \mathcal{N}(z \mid b, B), \qquad p(x \mid z) = \mathcal{N}\left(x \middle| a + \frac{K}{B}(z - b), A - \frac{K^2}{B} \right).$$

(a) Compute the prior and posterior predictive distributions $p(y)$ and $p(y \mid \mathcal{Y}_c)$ if the calibration set holds one reading, *i.e.* $\mathcal{Y}_c = \{y_o\}$.

Solution. We use the Gaussian identities of the statement.

We first identify the components we have:

- $p(y \mid \mu_y) = \mathcal{N}(y \mid \mu_y, \sigma^2)$
- $p(\mu_y) = \mathcal{N}(\mu_y \mid \mu_l, \sigma_l^2)$

Prior predictive

$$p(y) = \int p(y \mid \mu_y) p(\mu_y) d\mu_y.$$

We identify the parameters of the first identity as

$$\begin{aligned} x &= \mu_y \\ m &= \mu_l \\ S &= \sigma_l^2 \\ z &= y \\ h &= 1 \\ u &= 0 \\ R &= \sigma^2 \end{aligned}$$

which corresponds to the joint

$$p(\mu_y, y) = \mathcal{N}\left(\begin{pmatrix} \mu_y \\ y \end{pmatrix} \middle| \begin{pmatrix} \mu_l \\ \mu_l \end{pmatrix}, \begin{pmatrix} \sigma_l^2 & \sigma_l^2 \\ \sigma_l^2 & \sigma_l^2 + \sigma^2 \end{pmatrix} \right),$$

from which the second identity gives $p(y) = \mathcal{N}(y \mid \mu_l, \sigma^2 + \sigma_l^2)$.

Posterior predictive

$$p(y \mid \mathcal{Y}_c) = \int p(y \mid \mu_y) p(\mu_y \mid \mathcal{Y}_c) d\mu_y.$$

We first need the posterior $p(\mu_y \mid \mathcal{Y}_c)$. The joint of (μ_y, y_o) is the same as above, with $z = y_o$; the second identity, with $a = b = \mu_l$, $A = \sigma_l^2$, $B = \sigma_l^2 + \sigma^2$ and $K = \sigma_l^2$, gives

$$\begin{aligned} p(\mu_y \mid y_o) &= \mathcal{N}(\mu_y \mid \mu_{\text{post}}, \sigma_{\text{post}}^2) \\ &= \mathcal{N}\left(\mu_y \middle| \mu_l + \frac{\sigma_l^2}{\sigma^2 + \sigma_l^2}(y_o - \mu_l), \sigma_l^2 - \frac{\sigma_l^4}{\sigma^2 + \sigma_l^2} \right). \end{aligned}$$

After simplification, we identify the posterior parameters as

$$\mu_{\text{post}} = \frac{\sigma^2}{\sigma^2 + \sigma_l^2}\mu_l + \frac{\sigma_l^2}{\sigma^2 + \sigma_l^2} y_o$$

and

$$\sigma_{\text{post}}^2 = \frac{\sigma^2}{\sigma_l^2 + \sigma^2}\sigma_l^2.$$

Following the same development as for the prior predictive, we have

$$p(y \mid \mathcal{Y}_c) = \mathcal{N}(y \mid \mu_{\text{post}}, \sigma^2 + \sigma_{\text{post}}^2).$$

(b) Looking at the predictive uncertainty, what is the benefit of incorporating $\mathcal{Y}_c$?

Solution. As a reminder, we found

$$p(y) = \mathcal{N}(y \mid \mu_l, \sigma^2 + \sigma_l^2).$$

$$p(y \mid \mathcal{Y}_c) = \mathcal{N}(y \mid \mu_{\text{post}}, \sigma^2 + \sigma_{\text{post}}^2).$$

Comparing the variances, since $\sigma^2/(\sigma^2 + \sigma_l^2) \leq 1$,

$$\sigma_{\text{post}}^2 = \frac{\sigma^2}{\sigma^2 + \sigma_l^2}\sigma_l^2 \leq \sigma_l^2, \qquad \text{hence} \qquad \sigma^2 + \sigma_{\text{post}}^2 \leq \sigma^2 + \sigma_l^2.$$

The posterior predictive has a smaller or equal variance than the prior predictive. Incorporating the calibration set reduces the uncertainty about the next reading.

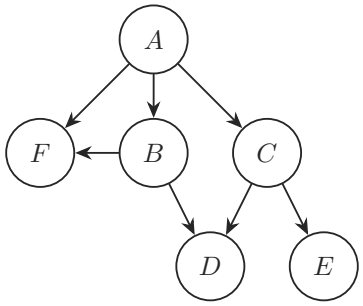
If we computed the variance for a calibration set of C observations, we would have found a posterior variance

$$\sigma^2_{\text{post}} = \frac{\sigma^2}{C\sigma_l^2 + \sigma^2}\sigma_l^2.$$

Since σ^2 is fixed, including more observations reduces the posterior predictive variance further. In the limit case $C \to \infty$, the posterior predictive variance converges to σ^2, the noise of the lidar itself, which no calibration can reduce (*aleatoric* uncertainty), while the uncertainty about μ_y vanishes.

Exercise 6 Independence and variable elimination

Which of the following statements does the structure of the network below guarantee?



Solution. We apply the d-separation algorithm. To show that two variables X and Y could be dependent, it is sufficient to find a single *active* (undirected) path from X to Y. A path is active if all of its consecutive triples are active.

(a) $\mathbf{P}(B, C) = \mathbf{P}(B)\mathbf{P}(C)$

Solution. This is true iff (if and only if) $B \perp C$ (B is independent of C). The three paths that link B and C are (B, A, C), (B, D, C) and (B, F, A, C). (B, A, C) is active because A is unobserved (common parent). We cannot guarantee that $B \perp C$.

(b) $\mathbf{P}(B, C \mid A) = \mathbf{P}(B \mid A)\mathbf{P}(C \mid A)$

Solution. This is true iff $B \perp C \mid A$. This time, (B, A, C) is not active because A is known. (B, D, C) is not active either because D is unobserved (v-structure). In (B, F, A, C), the triple (B, F, A) is not active (v-structure). (F, A, C) is not active either. We can guarantee that $B \perp C \mid A$.

(c) $\mathbf{P}(F, E) = \mathbf{P}(F)\mathbf{P}(E)$

Solution. This is true iff $F \perp E$. The four paths that link F and E are (F, A, C, E), (F, A, B, D, C, E), (F, B, A, C, E) and (F, B, D, C, E). In (F, A, C, E), the triples (F, A, C) (common parent) and (A, C, E) (cascade) are both active. We cannot guarantee that $F \perp E$.

(d) $\mathbf{P}(F \mid A, D, E) = \mathbf{P}(F \mid A, D)$

Solution. This is true iff $F \perp E \mid A, D$. This time, the triples (F, A, C), (F, A, B) and (B, A, C) are inactive (common parent, A observed). In (F, B, D, C, E), (F, B, D) is active (cascade, B unobserved). Indeed, observing an extremity of a triple does not change whether it is active: only the center variable matters, and, for a v-structure, its descendants. Then we have (B, D, C) which is active (v-structure, D observed) and (D, C, E) also. Hence, (F, B, D, C, E) is active and we cannot guarantee that $F \perp E \mid A, D$.

(e) $\mathbf{P}(B, E) = \sum_{a,c,d,f} P(a)\mathbf{P}(B \mid a)P(c \mid a)\mathbf{P}(d \mid B, c)\mathbf{P}(E \mid c)\mathbf{P}(f \mid a, B)$

Solution. The right-hand side is $\sum_{a,c,d,f} \mathbf{P}(a, B, c, d, E, f)$, with the joint distribution factored over the network as $\prod_i \mathbf{P}(X_i \mid \text{parents}(X_i))$. The statement always holds.

(f) Use inference by variable elimination to compute $\mathbf{P}(E \mid A = 1, B = 1)$.

Solution. The hidden variables are C, D and F. We have

$$\begin{aligned}
\mathbf{P}(E \mid A = 1, B = 1) &= \frac{1}{Z} \sum_{c,d,f} \mathbf{P}(A = 1, B = 1, c, d, E, f) \\
&= \frac{1}{Z} \sum_{c,d,f} P(A = 1)P(B = 1 \mid A = 1)P(c \mid A = 1) \\
&\quad P(d \mid B = 1, c)\mathbf{P}(E \mid c)P(f \mid A = 1, B = 1) \\
&= \frac{1}{Z} P(A = 1)P(B = 1 \mid A = 1) \sum_f P(f \mid A = 1, B = 1) \\
&\quad \sum_c P(c \mid A = 1)\mathbf{P}(E \mid c) \sum_d P(d \mid B = 1, c).
\end{aligned}$$

We define the initial factors as

$$\begin{aligned}
f_1 &= P(A = 1) \\
f_2 &= P(B = 1 \mid A = 1) \\
f_3(C) &= \mathbf{P}(C \mid A = 1) \\
f_4(C, E) &= \mathbf{P}(E \mid C) \\
f_5(C, D) &= \mathbf{P}(D \mid B = 1, C) \\
f_6(F) &= \mathbf{P}(F \mid A = 1, B = 1)
\end{aligned}$$

and the composite factors as

$$\begin{aligned}
f_7 &= \sum_f f_6(f) = 1 \\
f_8(C) &= \sum_d f_5(C, d) = 1 \\
f_9(C, E) &= f_3(C) \times f_4(C, E) \times f_8(C) \\
f_{10}(E) &= \sum_c f_9(c, E) \\
f_{11}(E) &= f_1 \times f_2 \times f_7 \times f_{10}(E).
\end{aligned}$$

Finally,

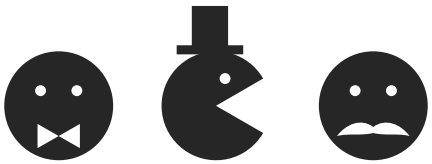
$$\mathbf{P}(E \mid A = 1, B = 1) = \frac{1}{Z} f_{11}(E) = \frac{f_{11}(E)}{\sum_e f_{11}(e)}.$$

Eliminating F and D gives factors equal to 1: a hidden variable that is not an ancestor of the query or the evidence sums out to 1 and can be removed from the network (the relevance theorem of lecture 5). The constant $f_1 f_2$ cancels in the normalization. What remains is

$$\mathbf{P}(E \mid A = 1, B = 1) = \sum_c P(c \mid A = 1)\mathbf{P}(E \mid c) = \mathbf{P}(E \mid A = 1),$$

already normalized. The evidence $B = 1$ does not matter: $E \perp B \mid A$ by d-separation, since every path from B to E reaches C through (B, A, C), (B, D, C) or (B, F, A, C), all inactive given A, as in (b).

Exercise 7 Recognizing Pacman (CS188, Spring 2014)
The ghosts of the maze are controlled by a program that watches the maze through cameras and must recognize Pacman in their pictures. It learns to do so from a set of pictures of Pacman, and of ghosts as negative examples.

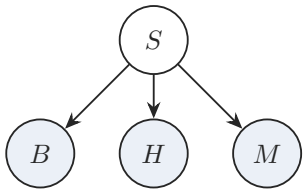


Because the pictures are black and white, and were taken from various angles, the program identifies Pacman from salient features: the presence of a bowtie B, hat H or mustache M. The following table summarizes the content of the pictures. Each feature takes values in $\{0, 1\}$, where 0 and 1 mean the feature is respectively absent and present. The subject of the picture is described by a random variable $S \in \{0, 1\}$, where 0 is a ghost and 1 is Pacman.

B	H	M	S
0	0	0	1
1	0	0	0
1	0	1	1
1	1	0	0
0	1	0	0
1	1	1	1

(a) Suppose the program is a naive Bayes classifier. Draw the Bayesian network that represents the dependencies between S, B, H and M for the program.

Solution.



(b) Write the Bayesian classification rule for this problem, *i.e.* the formula that given a data point (b, h, m) returns the most likely subject. Write the formula in terms of conditional and prior probabilities. What does the formula become under the assumptions of the program?

Solution. Given (b, h, m), the most likely subject is the one of highest posterior probability,

$$\begin{aligned} s^* &= \arg\max_s P(s \mid b, h, m) \\ &= \arg\max_s P(b, h, m \mid s) P(s). \end{aligned}$$

Under the naive Bayes assumptions of the program, B, H and M are conditionally independent given S, *i.e.* $\mathbf{P}(B, H, M \mid S) = \mathbf{P}(B \mid S)\mathbf{P}(H \mid S)\mathbf{P}(M \mid S)$. Then, the formula becomes

$$s^* = \arg\max_s P(b \mid s) P(h \mid s) P(m \mid s) P(s).$$

(c) What are the parameters of this model? Give estimates of these parameters according to the pictures of the table.

Solution. The parameters of the model are the elements of the prior vector $\mathbf{P}(S)$ and the (conditional) probability matrices $\mathbf{P}(B \mid S)$, $\mathbf{P}(H \mid S)$ and $\mathbf{P}(M \mid S)$. Their maximum likelihood estimates are the relative frequencies of the corresponding events among the pictures.

s	$P(S=s)$	$P(B=1 \mid S=s)$	$P(H=1 \mid S=s)$	$P(M=1 \mid S=s)$
0	$\frac{3}{6}$	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{0}{3}$
1	$\frac{3}{6}$	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{2}{3}$

The estimate $P(M=1 \mid S=0)=0$ is a problem: none of the three ghosts has a mustache, so any picture with a mustache gets the posterior probability 0 of being a ghost, whatever its other features. With so few pictures, a prior on the parameters helps. With a prior $\text{Beta}(\theta \mid 2,2) = \frac{1}{Z}\theta(1-\theta)$ on each parameter θ, the MAP estimates are $(k+1)/(n+2)$ for k occurrences among n pictures (*add-one smoothing*), *e.g.* $P(M=1 \mid S=0)=1/5$.

(d) Pacman eventually shows up in front of a camera wearing a bowtie, but no hat or mustache. Will the program recognize him?

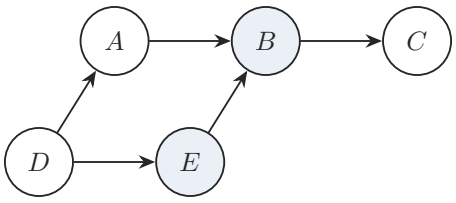
Solution. The program will recognize Pacman if $s^*=1$ for $(b,h,m)=(1,0,0)$. Using the parameters estimated previously, we have

$$P(b \mid 0)P(h \mid 0)P(m \mid 0)P(0) = \frac{2}{3} \times \left(1-\frac{2}{3}\right) \times \left(1-\frac{0}{3}\right) \times \frac{3}{6} \approx 0.111$$
$$P(b \mid 1)P(h \mid 1)P(m \mid 1)P(1) = \frac{2}{3} \times \left(1-\frac{1}{3}\right) \times \left(1-\frac{2}{3}\right) \times \frac{3}{6} \approx 0.074.$$

Therefore, $s^*=0$, meaning that the program will *not* recognize Pacman, who escapes.

Quiz

1. Consider the Bayesian network shown below. We want to infer $\mathbf{P}(A \mid b,e)$ where A is the query variable, B and E are evidence variables, and C and D are hidden variables. Which of the following statements is true?



- ○ $\mathbf{P}(A \mid b,e) \propto \sum_c \sum_d P(c \mid b)\mathbf{P}(b \mid A)P(b \mid e)\mathbf{P}(A \mid d)P(e \mid d)P(d)$
- ○ $\mathbf{P}(A \mid b,e) \propto \sum_c \sum_d P(c \mid b)\mathbf{P}(b \mid A,e)\mathbf{P}(A \mid d)P(e \mid d)$
- ● $\mathbf{P}(A \mid b,e) \propto \sum_c \sum_d P(c \mid b)\mathbf{P}(b \mid A,e)\mathbf{P}(A \mid d)P(e \mid d)P(d)$
- ○ $\mathbf{P}(A \mid b,e) \propto \sum_b \sum_e P(c \mid b)\mathbf{P}(A \mid b,e)\mathbf{P}(A \mid d)P(e \mid d)P(d)$

2. In a Bayesian network

- ● independence between two variables is guaranteed if all paths connecting them are inactive.
- ○ dependence between two variables can be guaranteed using d-separation.
- ○ a path between two nodes is active if it contains at least one active triple.
- ○ a cascade triple is active if the center node is observed.

3. Consider a Bayesian network with a single binary variable, following a Bernoulli distribution. If we observe T positive realizations and F negative ones,

   ◯ the maximum likelihood estimate of the positive probability is $\frac{T}{F}$.

   ◯ the maximum likelihood estimate of the positive probability is $\frac{T+F}{T}$.

   ◉ the maximum likelihood estimate of the positive probability is $\frac{T}{T+F}$.

   ◯ none of the above.